

Adaptive Traffic Signal Control by Choosing Phase Duration

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Abstract—An increase in the traffic flow intensity leads to increased transportation costs, inclusive of heightened travel duration and fuel consumption. Consequently, adaptive traffic control remains a central challenge in modern urban cities. Conversely, an increase in the number of data sources and their volume, alongside advancements in machine learning and artificial intelligence techniques, provide opportunities to enhance the efficiency of solving traffic control issues. One of the most common approaches to solving traffic control problems is adaptive traffic signal control at signalized intersections. To solve this problem, various control approaches are used, including choosing the next phase, keeping or changing the current phase, and choosing the duration of the next phase. This paper focuses on an adaptive traffic signal control problem denoted as adaptively selecting the duration of the next phase from a predefined sequence of phases. We propose a modified adaptive traffic signal method aimed at maximizing the weighted traffic flow passing through the intersection during the considered time interval (phase duration). Experimental studies conducted using the SUMO microscopic traffic simulation package validated the efficacy of our proposed approach.

Keywords—adaptive control, traffic signal control, traffic flow prediction, connected vehicles

I. INTRODUCTION

Effective traffic control in an urban environment remains an actual problem in transportation networks, the solution of which is necessary to increase urban mobility by reducing traffic congestion and travel time, increasing transport safety, and reducing the impact on the environment. Enhancing the efficiency of control methods can be achieved by using a heterogeneous data types detailing the intricate dynamics of traffic. This encompasses gathering data from various sources such as road sensors, loop detectors, and closed-circuit television (CCTV) cameras.

Currently, the development of connected vehicles (CV) and autonomous vehicles (AV) allows us to use information about the trajectory, position, and speed of such vehicles to solve different transportation problems, such as traffic flow prediction [1], motion control and trajectory tracking [2], navigation, collision avoidance [3] and so on. Utilizing adaptive traffic signal control within a connected vehicle environment, comprising both human-driven vehicles and connected vehicles (CV), emerges as a highly promising strategy for enhancing traffic control efficiency [4]. Connected vehicle technologies have shown an ability to alleviate congestion, minimize travel time and delay time [5], [6]. In [7], the authors concluded that even partial implementation of CV and AV can lead to enhanced efficiency and increased capacity at signalized intersections,

as opposed to scenarios assuming a homogeneous traffic flow consisting solely of human-driven vehicles.

Traditional traffic signal control systems, operating on fixed predetermined phase cycles, are not able to take into account the dynamic characteristics of traffic flows, resulting in congestion and increased travel times. To solve these problems, control systems with adaptive traffic signal control are widely employed. These systems use real-time information from vehicles, detectors and video surveillance systems to solve traffic flow control and optimization problems.

To develop adaptive traffic control systems, both classical control methods and machine learning techniques, particularly based on reinforcement learning, can be employed. Considering the traffic signal control system as a multi-agent system, it is necessary at each stage to determine the type and the set of actions accessible to each agent. In adaptive traffic signal control systems, the following types of systems can be distinguished [8]:

- In cycle-based methods, the determination of cycle length, phase order, and phase duration is performed at the end of a cycle.
- Conversely, in phase-based methods, the determination of phase duration and phase selection can be performed at the end of a phase.

In more detail, action design in phase-based systems includes:

- Choose the subsequent phase of a traffic signal from a set of predefined phases [9], [10].
- Deciding whether to keep or alter the current phase [11], [12].
- Choose the duration of the next phase [13], [14].

The next subsection provides an overview of the main traffic signal control methods.

The most popular approach in scientific works is adaptive control through the selection of the subsequent phase within the traffic signal cycle. This approach can significantly reduce waiting time at intersections in free-flow conditions. At the same time, using a traffic signal cycle structured as a predetermined sequence of phases and displaying the current phase duration is more convenient for drivers and more consistent with actual traffic conditions [15].

As a result, this paper addresses the adaptive traffic signal control problem as follows: adaptively select the duration of the subsequent phase from a predefined sequence of phases. To tackle this problem, we propose a modified approach aimed at maximizing the weighted traffic flow traversing the

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intersection during the considered time interval (phase duration) [9]. While the method proposed in [9] focuses on choosing the subsequent phase of a traffic signal, this paper introduces a modification of the method for selecting the duration of the next (predefined) phase.

II. LITERATURE REVIEW

The traffic signal control problem continues to be one of the most challenging issues in intelligent transportation systems. As a result, a significant number of scientific papers are dedicated to addressing this challenge.

Early approaches considered primarily rule-based methods and selected the next traffic signal phase from a predefined set according to a specified rule, such as fixed-time control [16] or actuated control [17]. More detailed descriptions of early methods can be found in the reviews [18], [19]. Currently, the active development of machine learning methods and artificial intelligence techniques leads to the extensive use of these methods to tackle the traffic signal control problem, notably through reinforcement learning (RL) approaches. In RL methods, an agent interacts with the environment, learning to make decisions based on these interactions.

The paper [20] provides an in-depth exploration of recent advancements in deep reinforcement learning techniques for addressing traffic signal control problems. The authors provided details on various problem formulations and the design of RL models, and discussed challenges and open questions. The authors emphasized that the most common action is to choose one of the possible green phases (actions) from a set of actions. Choosing a phase duration with a predetermined length is a relatively less common approach [21]. In [8], the authors presented a comprehensive literature review of 160 included articles by different criteria, including the type of the proposed control methods (RL/non-RL), components of the RL methods (state space definition, action space, reward), etc. A review of traffic control methods aimed at motivating for studying coordinated control of traffic signals and intelligent vehicles was presented in [22].

In the presented surveys [8], [20], the authors concluded that the most common approach to define an action is to select the subsequent traffic signal phase from a predetermined set of phases. In [23], the authors introduced a double deep Q network as a solution for tackling the traffic signal control problem at a single intersection. A deep RL model that uses a graph attention network to provide cooperation between agents was proposed in [10]. However, the authors considered the intersections of the same structure. In [24], the authors explored a scenario with a partially observable environment featuring connected vehicles. They proposed a deep Q-learning model designed to control a traffic signal at an isolated intersection within this environment. A deep Q-learning network to coordinate traffic lights was developed in [25]. Non-RL approaches to solve this task were presented in [9], [26], [27]. In [26], the authors calculated the queue of vehicles on both incoming and outgoing lanes and stabilized a "pressure" – the difference between outgoing and incoming traffic. Maximization of weighted traffic flow was proposed in [9], [27].

The "keep or change" strategy was considered in [11]. The authors developed a model concentrated on coordinating multiple intersections. A Q-network based approach for traffic control that used the same strategy was considered in [12].

However, the authors considered only two-phase traffic signals.

Actor-critic based adaptive traffic signal control was discussed in [13]. The authors considered a predefined sequence of phases and chose the duration of the current green phase, however, they considered the existence of a zero-duration phase as well. In [28], the authors designed an RL action that "selects a green time duration from a predefined set of durations" and evaluated the robustness of different RL models against different non-typical situations such as traffic incidents or jaywalking. A similar approach based on choosing the duration of the current phase was discussed in [14]. The authors proposed a universal neural network model to describe each agent for intersections with different structures and, as a consequence, different state and action spaces. The presented data augmentation method makes it possible to increase the efficiency of traffic control at new (unseen) intersections.

The presented review showed the active study of the RL methods for addressing the traffic signal control problem. However, there are still some limitations to use these methods in real-world scenarios due to their complexity. This paper introduces a non-reinforcement learning method for adaptive traffic signal control based on choosing the duration of the next (predefined) phase.

III. ADAPTIVE CONTROL

We consider a connected vehicle environment that includes connected vehicles, for which position, speed, and route are assumed to be known. The proposed approach consists of two main steps:

- Predict the travel time required for each vehicle to reach the nearest signalized intersection (subsection III.B).
- Using this information, choose the duration of the next traffic signal phase (subsection III.A).

Consider these steps in more detail.

A. Adaptive Traffic Signal Control Methods

To describe the proposed adaptive control algorithm, introduce the following notation. Let C be a set of connected vehicles, for each vehicle $c_i \in C$, it is assumed that the time interval t_i^l required to reach the intersection l and the current waiting time w_i are known. We denote the set of traffic signal phases as P , $p_{next} \in P$ is the next phase, d_{out} is a chosen duration of the next phase. We assume that the phase duration can take values from the predefined set $D = [5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60]$ (time in seconds).

Using the introduced notation, the proposed algorithm in a pseudocode can be described as presented in Algorithm 1. In the algorithm, $movement_allowed(c_i, p_{next}, d)$ is a function that determines whether a vehicle c_i passes through an intersection during a phase p_{next} with phase duration d . The main part of this function is to estimation the travel time needed for vehicles to traverse the intersection. To achieve this, an approach relying on artificial neural networks is employed for travel time estimation. (subsection III.B).

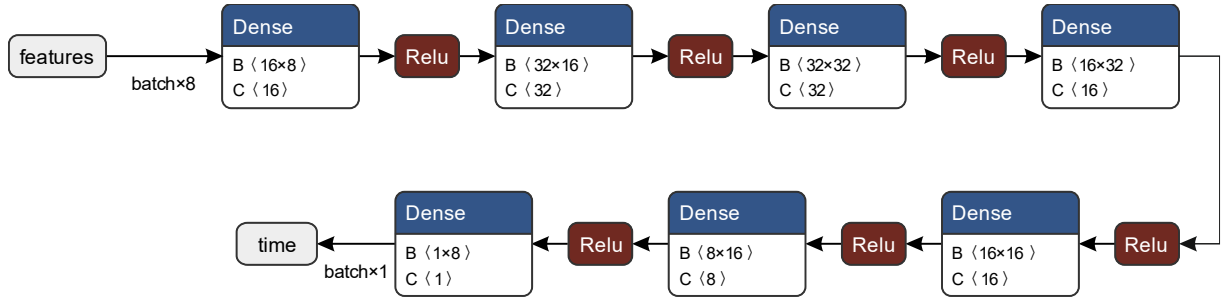


Fig. 1. The architecture of the neural network.

Algorithm 1:

Input data: C, D, p_{next}

Output data: d_{out}

$m = \text{zeros}(|D|)$ // array of zeros

for c_i **in** C :

for i **in** $\text{range}(|D|)$:

$d = D[i]$

if $\text{movement_allowed}(c_i, p_{next}, d)$:

$m[i] = m[i] + 1 + \alpha w_i$

else:

$m[i] = m[i] - \alpha(w_i + d)$

end for

end for

$i = \arg \max(m)$

$d_{out} = D[i]$

return d_{out}

Thus, the algorithm calculates the number of vehicles that will be able to pass the intersection during the chosen phase duration, taking into account the vehicle's waiting time, weighted by a coefficient α .

B. Travel Time Prediction Algorithm

In order to determine the number of vehicles traversing the intersection within the phase duration interval, it is necessary to forecast the travel time required to reach the intersection. In this paper, we propose to use a neural network-based algorithm [9]. The input feature vector of the model consists of the following parameters:

- The position, speed, and acceleration of the considered vehicle.
- The maximum speed on the current road lane.
- The number of vehicles preceding the vehicle under consideration.
- The type of turn required at the intersection.
- The position and speed of the first vehicle on the adjacent road lane.

The output value of the neural network is the predicted travel time. Fig. 1 shows the architecture of the neural network.

In the following section, we describe the experimental results of the developed approach.

IV. EXPERIMENTAL STUDY

The experimental studies of the proposed method were conducted using the SUMO microscopic simulation package [29] in a small real-world scenario with 8 signalized intersections [30]. We evaluate the effectiveness of the proposed method based on the adaptive selection of the predetermined phase duration with the following baseline approaches:

- Fixed duration – non-adaptive traffic signal control based on the selection of a fixed (predetermined) duration of each phase.
- MaxPWFlow control method [9] – a method based on adaptive selection of the next phase (rather than the phase duration).

The described methods were compared according to the criterion of average travel time and average waiting time at the intersection. To evaluate the effectiveness, we run 10 episodes to simulate vehicle movement in the morning rush hour. The comparison results are presented in Table I.

TABLE I. COMPARISON OF METHODS

<i>Model</i>	<i>Average waiting time, sec</i>	<i>Average travel time, sec</i>
Fixed duration	27.2	116.5
Proposed method	17.9	106.2
MaxPWFlow	3.2	88.3

Although the developed approach is inferior to the MaxPWFlow algorithm, it can reduce the average travel time and average waiting time compared to non-adaptive traffic signal control. Finally, we evaluate the travel time for each simulation episode. Fig. 2 illustrates a comparison of the methods based on the average travel time criterion.

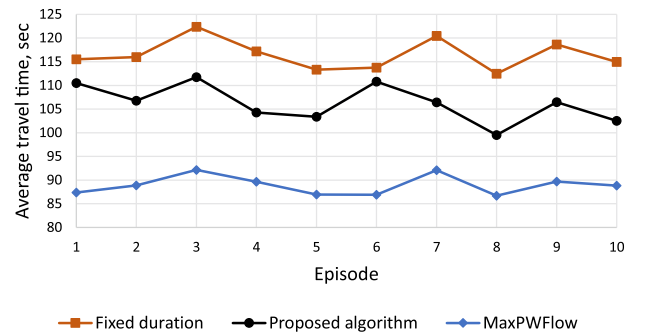


Fig. 2. Comparison by average travel time criterion.

The proposed method outperforms the non-adaptive approach in each episode. In addition, unlike the MaxPWFLOW algorithm, in the proposed method the sequence of phases remains constant, and their duration adaptively changes. The proposed approach can be applied in cases where it is necessary to use a predetermined cycle of phases. In addition, it is more convenient for drivers and pedestrians.

V. CONCLUSION

This paper addresses an adaptive traffic signal control problem by controlling the duration of the subsequent traffic signal phase. It is assumed that the phase order is fixed and the next phase is chosen from a predetermined sequence of phases. This approach differs from the more common approach to choosing the next phase, but the considered problem statement is more practical, convenient for drivers, and more consistent with actual traffic conditions.

To tackle the problem, we develop a modified adaptive traffic signal control method aimed at maximizing the weighted traffic flow. Experimental studies conducted using the SUMO microscopic traffic simulation package confirmed the effectiveness of the proposed approach.

In future studies, we plan to compare the proposed approach with more complex machine learning algorithms based on reinforcement learning models.

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